# A finite audit of hypergraph-product codes: supporting checks and unresolved dissertation findings

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**Abstract**

We could not establish the full dissertation findings of Quintavalle (2022), who reported oracle-assisted decoding guarantees, logical gate constructions, and confinement-based single-shot correction. We completed 0 of 4 sustainable thresholds, 0 of 2 capacity analyses, 0 of 2 suppression fits, and 0 of 3 random-product performance analyses. Finite support includes 1,061 guaranteed-range phase-error checks, 256 seed pairs, 19 matrix comparisons, 560,000 reconstructed trials and 80,000 original-software diagnostic trials. The thesis text was read and the rendered headline pages were inspected; finite tests do not prove universal results.

| Claim | Published | Ours | Effect magnitude | Direction | Verdict |
|---|---|---|---|---|---|
| ReShape guarantee Non-scalar | Corrects errors through (d-1)/2 using classical oracles. | 1,061 phase-error checks pass; no general proof. | not established | not established | partial |
| Toric scaling Non-scalar | Toric ReShape shows anti-threshold behavior. | p=.03: failure .0596 to .1828 as d=6 to 20; reconstructed. | not established | not established | partial |
| Random scaling Non-scalar | Random (3,4) scaling unclear; adverse large-distance trend expected. | Not run: original random seeds/grid not recovered. | not established | not established | unassessed |
| Logical basis Non-scalar | Every HGP has a symplectic logical line basis. | 256 seed pairs plus six examples pass; finite checks. | not established | not established | partial |
| Pieceable CZ Non-scalar | Round-robin logical CZ with intermediate ReShape correction. | 361 schedules and 256 ideal logical pairs pass; noisy steps untested. | not established | not established | partial |
| Adversarial noise Non-scalar | Good confinement and growing distance imply adversarial single-shot. | 31,232 residual checks on 12 qubits; no general proof. | not established | not established | partial |
| Stochastic noise Non-scalar | Good linear confinement implies stochastic single-shot threshold. | Not independently proved; theorem conditions inventoried. | not established | not established | unassessed |
| 3D confinement Non-scalar | 3D products have cubic X confinement; t is Z distance. | 4,096 errors enumerated on 12 qubits; no general proof. | not established | not established | partial |
| Sustainable thresholds Non-scalar | Surface 3.08(4)/2.90(1)%; toric 2.90(2)/2.78(2)%. | 0 of 4 sustainable-threshold estimates completed. | not established | not established | unassessed |
| Capacity contrast | BP+OSD 21.55(1)% versus cited sweep 15.5%. | 0 of 2 capacity fits completed; no fitted contrast. | not established | not established | unassessed |
| Repair subroutine Non-scalar | Repair improves high-noise decoding by over an order of magnitude. | 617 versus 72 failures/10000 at L=5, p=.1, q=.03, N=1. | not established | not established | partial |
| Suppression contrast | Beta 1.91(3) at N=0; 1.15(3) at N=8. | 0 of 2 suppression fits completed. | not established | not established | unassessed |
| Random products Non-scalar | Three random codes; crossing near 2.7% at N=16. | 0 of 3 performance analyses; original seeds not recovered. | not established | not established | unassessed |
| Open problems Non-scalar | Full gate framework open; general-confinement stochastic threshold conjectured. | No architecture or conjecture demonstration attempted. | not established | not established | unassessed |

Descriptive point estimates; not equivalence.

Table 1. Published: Quintavalle (2022), Chapters 3–6; exact locators in Appendix A. Ours: finite calculations and reconstructed runs. Non-scalar means qualitative or multicomponent; it does not erase either finding. "Unassessed" means the source finding was not independently assessed or its production fit was not completed. Threshold order is MWPM+BP/OSD, then BP/OSD twice; parentheses are source standard errors in the last digits. The confinement theorems require LDPC families with growing distance; the stochastic theorem also requires bounded qubit degree and good linear confinement.

Signed effects (Published, Ours): capacity 6.05 not established; suppression 0.76 not established. Capacity is BP+OSD minus cited sweep, in percentage points, tolerance 0.055; suppression is beta at N=0 minus N=8, dimensionless, tolerance 0.01. Positive means the first value exceeds the second. Rounding bounds define tolerances, not sampling uncertainty. No scalar aggregate is assigned to qualitative or multicomponent findings.

# 1 Introduction

We could not establish the full dissertation findings of Quintavalle (2022): the finite checks support parts of the decoding and gate constructions, but universal theorems and production estimates are unresolved. We completed 0 of 4 sustainable threshold estimates, 0 of 2 code-capacity analyses, 0 of 2 suppression fits, and 0 of 3 random-product performance analyses. The published sustainable thresholds are 3.08(4)% and 2.90(1)% for surface codes, and 2.90(2)% and 2.78(2)% for toric codes, in MWPM+BP/OSD and BP/OSD-twice order. There are no corresponding fitted values from this audit. Missing comparisons are not evidence of nonreplication.

The scientific question is which parts of the dissertation survive independent finite checks, and which require evidence beyond those checks. A hypergraph-product (HGP) quantum code is built from classical parity-check matrices; its parameters $[[n, k, d]]$ give physical qubits, logical qubits, and distance. Single-shot correction uses one noisy syndrome measurement per correction step. An assertion about an entire code family is stronger than an exact calculation on a small member of that family. The inventory in Appendix A retains all 41 claims across the six chapters and associated appendices; inventory completeness is not theorem verification.

# 2 What we did ourselves, and what we took from others

Our contribution is an independent finite implementation and comparison, not a new universal proof. We computed binary matrix ranks, kernels, logical bases, decoder outcomes and phase-sensitive gate images; built independent tensor and computational-basis checks; and ran reconstructed Monte Carlo experiments. The original work supplies the code definitions, conditional theorems, gate constructions, simulation designs and published numbers (Quintavalle, 2022). We distinguish the original software from our input settings: the former is pinned to an upstream revision, while missing source grids and depth choices make our runs diagnostic.

The full thesis PDF was read as text, and the rendered pages that support the headline and Published entries were inspected. Quotations follow that reading. Printed figures and equations on those pages were inspected. Sign conventions and equality-versus-bound readings that those figures do not settle remain provisional. Published theorem statements, numerical observations and conjectures are separate kinds of evidence throughout this paper.

# 3 Finite decoding and logical-operation findings

The decoding checks support the guaranteed finite regime and reproduce unfavorable toric scaling on a reconstructed grid. Chapter 3 Theorem 1 reported correction of errors of weight up to $(d-1)/2$ with classical seed and transposed-seed decoder oracles, $O(k)$ oracle calls and $O(n^2)$ additional operations (PDF p.44). The same article's abstract reported weight smaller than $(d-1)/2$. This audit follows the displayed theorem. The difference is not treated as an established error. Our 1,061 guaranteed-range phase-error checks all pass. Exhaustive classification includes 8,192 planar3 and 256 toric2 patterns; it is not a claim that all those patterns correct successfully. The canonical decomposition and logical line-basis checks pass for all 256 binary $2 \times 2$ seed pairs and six named examples.

The stochastic experiment supplies 560,000 reconstructed trials over 56 settings. At phase-flip probability .03, failure rates are .0596, .0623, .0714 and .1828 for distances 6, 8, 10 and 20. This is consistent with the source's toric anti-threshold observation (Chapter 3 Figure 4, PDF p.52), but does not establish an exact reproduction of its curves or tie conventions. The source describes random (3,4) scaling as less clear: the pseudo-threshold appears to increase, there is no common crossing, and a toric-like adverse trend is expected at larger distances (PDF p.52, lines 2644–2649). We did not run this family because its original seeds and grid were not recovered. Results and intervals are in `results/toric_noise.json`; the decoder and interval functions are `analysis/toric_noise.py` and `analysis/toric_noise.py`.

Logical basis and ideal gate checks support selected finite constructions, while noisy fault tolerance is untested. Square Hadamard–SWAP and square CZ–S pass phase-sensitive checks on planar3, planar5, toric2, toric3, Hamming65 and Hamming98. The latter two codes have 17 and 32 logical qubits. All nine symmetric-code appendix rows agree in construction parameters; quantum distance uses the source theorem rather than an independent exhaustive proof. There are 361 collision-free round-robin schedules and 256 successful ideal logical CZ pair actions. The source's intermediate ReShape correction (Chapter 4, Section 4.1, PDF pp.70–71) is a condition of the fault-tolerance construction, not something established by these ideal checks. Conventional ideal S and T injection gadgets pass both branches with conditional phase correction. The rendered round-robin figure (PDF p.71) and the S/T injection circuits (PDF p.72) were inspected; noisy injection was not run. Evidence is in `results/exact_checks.json` and `results/independent_checks.json`.

# 4 Confinement and single-shot findings

The finite confinement checks do not prove a stochastic threshold. Chapter 5 Theorem 1 requires a quantum low-density parity-check (LDPC) family with good confinement and distance $d \geq an^b$ for positive $a, b$. It establishes adversarial single-shot correction; good X confinement yields the phase-Z statement. Theorem 2 additionally requires bounded qubit degree and good linear confinement for its local-stochastic threshold. Theorem 3 states X confinement with $t$ equal to Z distance and $f(x) = x^3/4$ or better for 3D products (PDF p.86). Our 12-qubit finite example enumerates 4,096 physical errors and passes 31,232 residual checks. We did not independently re-prove these theorems. The proposed extension from general good confinement to a local-stochastic threshold is explicitly conjectural, as is clear from the text preceding Theorem 2.

The independent tensor construction matches all 19 available archived Hx/metacheck matrices entry for entry. This validates finite matrix construction, not the absent production fits. The original BP+OSD twice decoder (belief propagation followed by ordered-statistics decoding at both stages) ran eight diagnostic settings of 10,000 trials each. At $L = 5$, data noise $p = .1$, syndrome noise $q = .03$ and reconstructed depth $N = 1$, repair reduces failures from 617 to 72 per 10,000. This supports improvement in our diagnostic; the source's over-an-order-of-magnitude improvement refers to its high-syndrome-noise experiment (Chapter 5 Figure 2, PDF p.94), whose exact depth/grid was not recovered. Our final round injects fresh data noise before perfect measurement. `results/original_diagnostic_grid.json` records these outcomes and settings.

The production comparisons are unavailable. Chapter 5 Table I reports a toric phase-flip capacity threshold of 21.55(1)% for BP+OSD versus a cited sweep threshold of 15.5%; the 6.05 percentage-point contrast is a source comparison, not our experiment. Appendix E reports suppression exponents 1.91(3) at $N = 0$ and 1.15(3) at $N = 8$, a difference of .76. Table III reported $[[1336, 4, 6]]$, $[[3100, 5, 8]]$ and $[[5964, 6, 10]]$ random-product codes. A sentence on the same page reported that the distance-10 instance encodes ten logical qubits; the Published cell follows the table, and the difference is not treated as an established error. Figure 4 reported a suggested threshold at 2.7% after 16 rounds (PDF pp.95–96). None of those three performance analyses was completed. The sustainable-threshold parent retains its surface and toric contrasts, .18 and .12 percentage points, each with a .01 rounding tolerance; no pooled effect is justified.

# 5 Comparison precision and uncertainty

Finite algebraic checks have exact arithmetic but restricted scope. For stochastic trials we use Wilson intervals (Wilson, 1927) and a 56-cell Bonferroni adjustment (NIST/SEMATECH, n.d.); Wilson coverage is approximate, so the adjustment does not create an exact finite-sample family guarantee. No interval measures uncertainty from a reconstructed grid or from conventions the inspected pages do not settle. The source standard errors remain attached to its thresholds and exponents; they are not tolerance bounds. Method metadata and its verification limits are in `sources/method-citations.json`.

| Claim | Published | Ours | Precision | Numerical | Scope |
|---|---|---|---|---|---|
| Sustainable thresholds | 3.08/2.90; 2.90/2.78% | 0 of 4 fits | .01% | withheld | unavailable |
| Capacity contrast | 6.05 pp | no fit | .055 pp bound | withheld | unavailable |
| Suppression contrast | .76 | no fit | .01 bound | withheld | unavailable |
| Appendix parameters | 9 rows | 9 rows agree | integer | match | limited |
| Matrix construction | 19 archives | 19 agree | binary | match | limited |

The first three numerical verdicts are withheld because no comparable fitted result exists. Appendix parameter agreement excludes exhaustive quantum-distance certification. Matrix agreement concerns only the available finite archives. "Match" means equality at the printed precision (half-up rounding for decimals); "close" would mean at most one last-place unit without matching the rounding, and "mismatch" would mean a larger difference. These labels are not assigned to absent fits or nonnumeric theorems. We found no independently verified typographical correction; the candidate gate differences below are substantive but unresolved.

# 6 Assumptions

## 6.1 Reconstructed grid and decoder ties

We supplied a 56-setting toric grid with 10,000 trials per point and random seed 20261103. Distances 6, 8, 10 and 20 follow the identifiable source design; exact probabilities, coordinate complements and tie conventions are reconstructed. Half the draws use uniform random numbers and half binomial draws. Independent decoder agreement is checked away from ties, and valid syndrome recovery is checked throughout. The scope is recorded in `results/toric_noise.json`.

## 6.2 Diagnostic depth and source software

We supplied depth $N = 1$ for the eight repair diagnostics because the exact Figure 2 setup was unavailable. Four pilot conditions test feasibility rather than estimate thresholds. We retain the original final-round data-noise behavior and separate metacode failures from logical failures. These choices prevent the diagnostics from being counted as source-equivalent fits.

## 6.3 Rounding tolerances

We use the sum of source endpoint half-units for derived contrasts: .055 percentage points for capacity and .01 for suppression. Each sustainable-threshold topology contrast has .01 percentage-point tolerance. These are fixed source-resolution bounds, not margins inferred from our outcomes; no comparison can be resolved without corresponding estimates. Non-scalar claims have no fabricated numerical tolerance.

# 7 Ambiguities that were resolved

## 7.1 Independent matrix ordering

The tensor construction agrees with all 19 archived Hx/metacheck matrices, resolving the finite ordering comparison for the available inputs. `results/independent_checks.json` contains each topology, size and match indicator; `analysis/independent_checks.py` implements the independent tensor construction. This does not recover absent surface-L11 inputs.

## 7.2 Support versus phase in sibling CZ

Independent computational-basis phase checks reproduce negative stabilizer images from the symplectic calculation. Hamming98 has 16 negative images; toric2 also exhibits phase failures. A supplied Z Pauli correction restores the tested signs, but changes the gate and does not replicate the literal source expression. `analysis/independent_checks.py` and `results/independent_checks.json` document this distinction. Whether the source uses an implicit Pauli frame is unresolved.

# 8 What remains unresolved

The main unresolved comparisons require source inputs or proofs, rather than more interpretations of the finite successes. The following six limits delimit the evidence.

**Limit 1.** The thesis PDF was read and the rendered headline pages, including Chapter 4 figures on PDF pages 67–72, were inspected. Phase conventions and missing numerical grids remain provisional.

**Limit 2.** Finite binary checks do not prove the dissertation universal theorems or noisy fault tolerance.

**Limit 3.** Original stochastic grids, tie conventions, random seeds, MWPM implementation and surface L11 code were not recovered.

**Limit 4.** 0 of 4 sustainable thresholds, 0 of 2 capacity analyses, 0 of 2 suppression fits, and 0 of 3 random-code performance analyses were completed.

**Limit 5.** Toric L11 pilot completed 9 trials in 180 seconds: approximately 55.6 CPU-hours per 10000-trial point; full production grids are unresolved.

**Limit 6.** Sibling gate and partition discrepancies are candidate source-comparison gaps. The printed figures were inspected; the phase convention is not settled by those figures. They are not established source errors.

The candidate gate discrepancies have several causes still in contention: extraction or convention differences, implementation error, and an error in the source statement. The independent phase representation reduces concern about a single phase implementation, while truly symmetric matrices pass sibling Hadamard–SWAP checks that fail 18 support checks for weak image symmetry on toric3. Exact Z-sector partition distances 3, 2 and 3 for planar3, toric2 and toric3 differ from a literal extracted floor-half-distance expression. The evidence does not rank a source error above an unsettled phase convention. The equation images were inspected. Explicit symmetry and Pauli-frame conventions remain the highest-priority discriminating artifacts; X-sector and universal partition statements remain unverified.

The numerical stopping evidence is concrete: the toric-L11 pilot completed nine trials in 180 seconds, projecting about 55.6 CPU-hours per 10,000-trial point before complete grids. Surface-L11 inputs, the MWPM implementation, exact stochastic grids and random seeds were not recovered. `results/original_pilots.json` gives each attempted condition and outcome. Resource limits explain uncompleted estimates but do not validate them. There is no complete gate architecture demonstration, and nonlinear confinement must not be conflated with the proved conditional linear-confinement theorem.

# 9 Code and source availability

The dated availability ledger (3 November 2026) is `availability.md`. The route record is `provenance/source-acquisition.json`. Source text is `sources/browser-extraction.txt`, and the finding-to-source audit is `sources/headline-finding-audit.json`. The thesis PDF was obtained from the public record, read in full, and inspected on the rendered headline pages. The PDF file is not stored in this repository. Vendored MIT-licensed source is the unmodified single-shot repository at revision bdfb437b2abcfa514997f26be97a711b878448cb, with BP+OSD revision 74f86d3ef00f04bbb90a043dfef52e92a091f4d3 (Single-shot software, archived revision). `provenance/upstream.json` records the pins. Repository master links are navigation aids, not immutable reproduction identifiers.

# 10 Next steps

The thesis PDF and the rendered headline pages have been read, including the Chapter 4 figures. The phase convention is still unresolved. Next obtain the original random matrices, surface-L11 files, probability grids, seeds and MWPM code; validate their construction against the finite reference and rerun the planned threshold and exponent fits. A larger compute allocation without these inputs would still leave source comparability unresolved. Finally independently audit the full universal proofs and intermediate noisy correction before interpreting finite gate actions as fault tolerance.

# A Complete claim inventory

Every item below distinguishes its source attribution from our outcome and scope. All locators refer to Quintavalle (2022); PDF page numbers are one-based. The concise page-1 findings are expanded here rather than replaced by a pooled statistic.

## A.1 Context and notation

Claim ch1. Source: Ch1. Published: Context and notation.

Ours: Documentary inventory only; no independent proof, resource-count measurement or fit completed for this item. Status: not established. Documentary inventory only; no independent proof, resource-count measurement or fit completed for this item.

Required scope: Audit background and explicitly exclude previously published introductory results.

Evidence: sources/browser-extraction.txt, availability.md.

## A.2 Motivation and overview

Claim ch2. Source: Ch2. Published: Motivation and overview.

Ours: Documentary inventory only; no independent proof, resource-count measurement or fit completed for this item. Status: not established. Documentary inventory only; no independent proof, resource-count measurement or fit completed for this item.

Required scope: Map the three research questions to Chapters 3–5; do not treat architectural motivation as a demonstrated device.

Evidence: sources/browser-extraction.txt, availability.md.

## A.3 ReShape decoding

Claim ch3. Source: Ch3. Published: ReShape decoding.

Ours: Chapter-wide result not established; limited supporting results and blocked components are itemized separately in this inventory.. Status: not established. Chapter accounting is not a replication result.

Required scope: Cover canonical form, homological invariants, adversarial correctness, complexity, worked examples, and both stochastic-noise families.

Evidence: sources/browser-extraction.txt, availability.md.

## A.4 Logical gates

Claim ch4. Source: Ch4. Published: Logical gates.

Ours: Chapter-wide result not established; limited supporting results and blocked components are itemized separately in this inventory.. Status: not established. Chapter accounting is not a replication result.

Required scope: Cover canonical line bases, partition distances, square and symmetric gates, examples, pieceable circuits and injection.

Evidence: sources/browser-extraction.txt, availability.md.

## A.5 Confinement and single-shot correction

Claim ch5. Source: Ch5. Published: Confinement and single-shot correction.

Ours: Chapter-wide result not established; limited supporting results and blocked components are itemized separately in this inventory.. Status: not established. Chapter accounting is not a replication result.

Required scope: Cover theoretical results and all numerical analyses, including negative and unavailable outcomes.

Evidence: sources/browser-extraction.txt, availability.md.

## A.6 Outlook

Claim ch6. Source: Ch6. Published: Outlook.

Ours: Documentary inventory only; no independent proof, resource-count measurement or fit completed for this item. Status: not established. Documentary inventory only; no independent proof, resource-count measurement or fit completed for this item.

Required scope: Preserve open problems as open; distinguish conjectured extension to nonlinear confinement from established theorems.

Evidence: sources/browser-extraction.txt, availability.md.

## A.7 Canonical decomposition and quotient invariant

Claim reshape-canonical. Source: Dissertation: Canonical decomposition and quotient invariant. Published: Canonical decomposition and quotient invariant.

Ours: For all256 binary2x2 seed pairs and six named examples, structural assertions verify CSS commutation, ranks, complete symplectic line pairing and complement dimensions; no independent universal proof audit completed.. Status: partly replicated. Finite result stated explicitly; source-wide generalization or visual wiring not established.

Required scope: Implement GF(2) splitting and independently check stabilizer invariance and kernel representatives; audit Chapter 3 Propositions 1–3 and Appendices A–C.

Evidence: sources/browser-extraction.txt, availability.md.

## A.8 Adversarial decoding and shaped errors

Claim reshape-correctness. Source: Ch3 Theorem1; PDF44, L1764–1768@P43. Published: Any [[n,k,d]] HGP constructed from the classical seed parity checks can decode errors through (d-1)/2 with O(k) oracle calls to classical seed/transposed-seed decoders plus $O(n^2)$ operations; source theorem, not our finite proof..

Ours: 1061 guaranteed-range errors pass; 8448 exhaustive patterns classified. Status: partly replicated. Finite or documentary coverage only; universal and full-design conclusions not independently established. See named artifacts and limitations.

Required scope: Binary hypergraph product code [[n,k,d]]; suitable classical decoders for each seed and its transpose; adversarial errors through (d-1)/2. Our checks address phase errors and finite instances.

Evidence: sources/browser-extraction.txt, availability.md, results/exact_checks.json.

## A.9 Oracle and classical resource bounds

Claim reshape-cost. Source: Dissertation: Oracle and classical resource bounds. Published: Oracle and classical resource bounds.

Ours: Documentary inventory only; no independent proof, resource-count measurement or fit completed for this item. Status: not established. Documentary inventory only; no independent proof, resource-count measurement or fit completed for this item.

Required scope: Audit dimensions and count active oracle calls against logical dimension; distinguish operation counts from measured wall time.

Evidence: sources/browser-extraction.txt, availability.md.

## A.10 Figures 1–3 illustrative constructions

Claim reshape-examples. Source: Dissertation: Figures 1–3 illustrative constructions. Published: Figures 1–3 illustrative constructions.

Ours: Repetition-based planar3/5 and toric2/3 constructions checked algebraically; 8192 planar3 and256 toric2 errors classified. Image-only worked examples not independently visually verified.. Status: partly replicated. Finite result stated explicitly; source-wide generalization or visual wiring not established.

Required scope: Reconstruct repetition, toric and worked planar code structure and enumerate planar errors; preserve graphical input ambiguity if figures cannot be rendered.

Evidence: sources/browser-extraction.txt, availability.md.

## A.11 Figure 4 toric anti-threshold

Claim reshape-toric. Source: Ch3 Figure4; PDF52, L2612–2645@P51. Published: Toric ReShape performance worsens with size at substantial stochastic noise.

Ours: 560000 trials on reconstructed 56-setting grid; at p=.03 failure .0596,.0623,.0714,.1828 for L6,8,10,20. Status: partly replicated. Finite or documentary coverage only; universal and full-design conclusions not independently established. See named artifacts and limitations.

Required scope: Source toric HGP family under independent phase-flip noise; source Figure4 uses at least10000trials per point. Our reconstructed grid uses d=6,8,10,20 and14 probabilities; original point/tie conventions unrecovered.

Evidence: sources/browser-extraction.txt, availability.md, results/toric_noise.json.

## A.12 Figure 4 random (3,4) family

Claim reshape-random. Source: Ch3 Figure4 and discussion; PDF52, L2612–2614 and L2644–2649@P51. Published: Figure4 labels the random (3,4) family anti-threshold, but its discussion says the panel is less clear: pseudo-threshold appears to increase with distance and no common crossing is seen. The authors expect the toric trend for larger distances; that expectation is not a demonstrated asymptotic result..

Ours: Not run: original random seeds/grid not recovered.. Status: blocked. Not run: original random seeds/grid not recovered.

Required scope: Source HGP C(delta,delta transpose), full-rank randomly generated(3,4)seed matrices, independent phase-flip noise; original matrices and grid not recovered. Source discussion explicitly qualifies the random panel as less clear and expects, rather than demonstrates, the larger-distance toric trend.

Evidence: sources/browser-extraction.txt, availability.md.

## A.13 Canonical logical line basis

Claim line-basis. Source: Ch4 Theorem1; PDF63, L3598–3624@P62. Published: Canonical paired line-supported logical basis.

Ours: All 256 seed pairs and named examples satisfy finite basis checks. Status: partly replicated. Finite or documentary coverage only; universal and full-design conclusions not independently established. See named artifacts and limitations.

Required scope: Arbitrary binary hypergraph product code; logical X/Z line support with symplectic pairing. Our census covers256 ordered pairs of binary2x2seeds and six examples.

Evidence: sources/browser-extraction.txt, availability.md, results/exact_checks.json.

## A.14 Diagonal-twin partition distance

Claim diagonal-partition. Source: Ch4 diagonal-twin partition. Published: Extracted floor(d/2) partition-distance expression.

Ours: Z-sector distances 3,2,3 on planar3,toric2,toric3; equality versus bound remains unresolved after inspection of the partition text. Status: not established. Finite or documentary coverage only; universal and full-design conclusions not independently established. See named artifacts and limitations.

Required scope: Compute exact partition distances on tractable planar and toric instances; compare lower-bound versus equality wording and inspect rendered equations before alleging discrepancy.

Evidence: sources/browser-extraction.txt, availability.md, results/independent_checks.json.

## A.15 Square Hadamard-SWAP

Claim square-hswap. Source: Dissertation: Square Hadamard-SWAP. Published: Square Hadamard-SWAP.

Ours: All stabilizer generators and logical X/Z images pass phase-sensitive square H-SWAP assertions for planar3,planar5,toric2,toric3,Ham
No universal proof certified.. Status: partly replicated. Finite result stated explicitly; source-wide generalization or visual wiring not established.

Required scope: Verify full stabilizer and logical Pauli action on repetition and the specified 65-qubit Hamming example.

Evidence: sources/browser-extraction.txt, availability.md.

## A.16 Square CZ-S

Claim square-czs. Source: Dissertation: Square CZ-S. Published: Square CZ-S.

Ours: All stabilizer generators and logical X/Z images pass square CZ-S checks in the six named cases, including right-sector S dagger phases. No universal proof certified.. Status: partly replicated. Finite result stated explicitly; source-wide generalization or visual wiring not established.

Required scope: Verify phases as well as binary support for stabilizer preservation and logical action, including right-sector S dagger.

Evidence: sources/browser-extraction.txt, availability.md.

## A.17 Symmetric sibling partition distance

Claim sibling-partition. Source: Dissertation: Symmetric sibling partition distance. Published: Symmetric sibling partition distance.

Ours: Exact finite Z-sector sibling distances computed for toric2 and toric3; X-sector and general partition theorem not certified. Values in independent_checks.json partition_distances.. Status: partly replicated. Finite result stated explicitly; source-wide generalization or visual wiring not established.

Required scope: Verify exact distances on small symmetric examples and inspect sector-transversal argument.

Evidence: sources/browser-extraction.txt, availability.md.

## A.18 Sibling Hadamard-SWAP

Claim sibling-hswap. Source: Ch4 sibling Hadamard-SWAP proposition. Published: Sibling H-SWAP under symmetry condition.

Ours: Weak image symmetry toric3 fails 18 support checks; truly symmetric matrices pass; pending source review. Status: not established. Finite or documentary coverage only; universal and full-design conclusions not independently established. See named artifacts and limitations.

Required scope: Verify Pauli action on cyclic repetition and symmetric Hamming examples.

Evidence: sources/browser-extraction.txt, availability.md, results/exact_checks.json.

## A.19 Sibling CZ

Claim sibling-cz. Source: Ch4 sibling CZ proposition. Published: Sibling CZ preserves encoded space.

Ours: 16 negative stabilizer images in Hamming98, confirmed by independent basis phases; the printed circuits were inspected and the phase convention remains unresolved. Status: not established. Finite or documentary coverage only; universal and full-design conclusions not independently established. See named artifacts and limitations.

Required scope: Verify stabilizer preservation and logical CZ pairing on source symmetric examples.

Evidence: sources/browser-extraction.txt, availability.md, results/independent_checks.json.

## A.20 Hamming and toric examples

Claim gate-examples. Source: Dissertation: Hamming and toric examples. Published: Hamming and toric examples.

Ours: 65-qubit code has17 logical qubits;98-qubit code has32. Exact algebra verifies line bases; source gate failures separately retained. All9 appendix parameter rows computed; no exhaustive quantum-distance certification.. Status: partly replicated. Finite result stated explicitly; source-wide generalization or visual wiring not established.

Required scope: Reconstruct the specified seed matrices, code dimensions and logical basis, including 65 and 98 physical qubits.

Evidence: sources/browser-extraction.txt, availability.md.

## A.21 Round-robin entangling gates

Claim pieceable. Source: Ch4 Sec4.1; PDF70–71, L4186–4225@P69–70; limitations L4350–4363@P72. Published: Pieceable round-robin logical entangling gates.

Ours: 361 schedules and 256 ideal logical pair actions pass; noisy intermediate correction untested. Status: partly replicated. Finite or documentary coverage only; universal and full-design conclusions not independently established. See named artifacts and limitations.

Required scope: Logical CZ between left/right logical qubits using sector-transversal pieces and intermediate ReShape correction. High-weight intermediate X stabilizers and time overhead limit noisy fault-tolerance claims; our checks are ideal.

Evidence: sources/browser-extraction.txt, availability.md, results/independent_checks.json.

## A.22 Single-qubit injection and universality

Claim injection. Source: Dissertation: Single-qubit injection and universality. Published: Single-qubit injection and universality.

Ours: Conventional ideal S and T injection gadgets each pass two branch checks with conditional phase corrections. Source image-only wiring and noisy injection not verified; this is a related diagnostic.. Status: partly replicated. Finite result stated explicitly; source-wide generalization or visual wiring not established.

Required scope: Check ideal state-injection branches with exact small complex matrices; do not infer noisy gate thresholds or resource savings.

Evidence: sources/browser-extraction.txt, availability.md.

## A.23 Confinement implies adversarial single-shot correction

Claim confinement-adversarial. Source: Ch5 Theorem1; PDF86, L5046–5050@P85; definition L5040–5044. Published: An LDPC code family with good confinement and distance $d \geq an^b$ $(a, b > 0)$ is single shot for adversarial noise; good X confinement gives the phase-Z version..

Ours: 31232 finite syndrome/noise checks on 12-qubit surface instance pass. Status: partly replicated. Finite or documentary coverage only; universal and full-design conclusions not independently established. See named artifacts and limitations.

Required scope: Quantum LDPC family with good confinement and growing distance $d \geq an^b$, $a, b > 0$; good X confinement restricts guarantee to Pauli-Z noise. Our12qubit check is finite.

Evidence: sources/browser-extraction.txt, availability.md, results/exact_checks.json.

## A.24 Linear confinement implies stochastic single-shot threshold

Claim confinement-stochastic. Source: Ch5 Theorem2; PDF86, L5057–5061@P85; bounded qubit degree and polynomially growing distance. Published: An LDPC family with bounded qubit degree, good linear confinement and $d \geq an^b$ $(a, b > 0)$ has a sustainable single-shot threshold for local stochastic noise; good X confinement gives the phase-Z version..

Ours: Not independently proved; theorem conditions inventoried.. Status: not established. Not independently proved; theorem conditions inventoried.

Required scope: Quantum LDPC family with bounded qubit degree, good linear confinement, and $d \geq an^b$, $a, b > 0$; local stochastic noise. Good X confinement gives Pauli-Z version; general confinement extension is a conjecture.

Evidence: sources/browser-extraction.txt, availability.md.

## A.25 Cubic confinement of 3D products

Claim confinement-product. Source: Ch5 Theorem3; PDF86, L5062–5064@P85; $f(x) = x^3/4$ or better. Published: All 3D product codes have (t,f) X confinement with t equal to Z distance and $f(x) = x^3/4$ or better..

Ours: 4096 finite error words checked; universal proof not reproduced. Status: partly replicated. Finite or documentary coverage only; universal and full-design conclusions not independently established. See named artifacts and limitations.

Required scope: Three-dimensional product codes; X confinement t equals Z distance; $f(x) = x^3/4$ or better. Finite quotient check does not reproduce universal proof.

Evidence: sources/browser-extraction.txt, availability.md, results/exact_checks.json.

## A.26 Soundness and expander confinement

Claim confinement-expander. Source: Dissertation: Soundness and expander confinement. Published: Soundness and expander confinement.

Ours: Documentary inventory only; no independent proof, resource-count measurement or fit completed for this item. Status: not established. Documentary inventory only; no independent proof, resource-count measurement or fit completed for this item.

Required scope: Audit Lemma 2 and Appendix D, including strictness and expansion assumptions; no replacement by an unverified random graph.

Evidence: sources/browser-extraction.txt, availability.md.

## A.27 3D chain complex and locality

Claim product-construction. Source: Ch5 Appendix B. Published: 3D tensor chain complexes and metachecks.

Ours: All 19 available original Hx/metacheck matrices match independent construction entry for entry. Status: partly replicated. Finite or documentary coverage only; universal and full-design conclusions not independently established. See named artifacts and limitations.

Required scope: Independently build all three boundary maps, verify chain identities, rank parameters and metachecks against archived original matrices for available sizes; account for Appendix B.

Evidence: sources/browser-extraction.txt, availability.md, results/independent_checks.json.

## A.28 Two-stage decoding and invalid syndromes

Claim single-shot-algorithm. Source: Dissertation: Two-stage decoding and invalid syndromes. Published: Two-stage decoding and invalid syndromes.

Ours: Pinned original BP+OSD twice decoder built and executed in4 pilots and8 reconstructed diagnostic conditions of10000trials each. Independent tensor check matches19 original matrices; exact source production design not recovered.. Status: partly replicated. Finite result stated explicitly; source-wide generalization or visual wiring not established.

Required scope: Build the original C++ source with pinned dependency; validate metacode failure detection and compare independent construction and controls.

Evidence: sources/browser-extraction.txt, availability.md.

## A.29 Four sustainable threshold estimates

Claim threshold-table. Source: Ch5 TableII; PDF85, L4949–4955@P84; standard errors in parentheses; MWPM+BP/OSD versus BP/OSD twice. Published: Sustainable thresholds: surface MWPM/BPOSD 3.08(.04)%, BPOSD twice 2.90(.01)%; toric 2.90(.02)%, 2.78(.02)%.

Ours: 0 of 4 sustainable-threshold estimates completed.. Status: not established. 0 of 4 sustainable-threshold estimates completed.

Required scope: Source independent phase flips and measurement flips with q=p; sustainable large-cycle estimates for surface/toric and MWPM+BP/OSD versus BP/OSD twice. Parent contains four estimates, not one pooled scalar.

Evidence: sources/browser-extraction.txt, availability.md, results/original_pilots.json.

## A.30 Code-capacity threshold and prior-decoder comparison

Claim capacity. Source: Ch5 TableI; PDF85, L4893–4907@P84; phase-flip threshold, cited sweep comparator. Published: Toric BP+OSD 21.55(.01)% versus literature sweep 15.5%; surface approximately 21.6%.

Ours: 0 of 2 capacity fits completed; no fitted contrast.. Status: not established. 0 of 2 capacity fits completed; no fitted contrast.

Required scope: Toric phase-flip code-capacity BP+OSD estimate21.55(1)%;15.5%sweep value is cited prior work, not a new sweep experiment. Planned replication includes both toric and surface capacity analyses.

Evidence: sources/browser-extraction.txt, availability.md, results/original_pilots.json.

## A.31 Figure 2 failure-mode improvement

Claim failure-subroutine. Source: Ch5 Figure2 and text; PDF94, L5939@P93; high syndrome noise. Published: Syndrome failure repair improves logical performance at high measurement noise.

Ours: At q=.03, 617 versus 72 failures in 10000 trials each; reconstructed depth N1. Status: partly replicated. Finite or documentary coverage only; universal and full-design conclusions not independently established. See named artifacts and limitations.

Required scope: Source toric L=5,data p=.1 and varying syndrome noise; original figure depth/grid not recovered. Our diagnostic uses N=1 and original software final perfect-measurement round with fresh data noise,10000trials per arm.

Evidence: sources/browser-extraction.txt, availability.md, results/original_diagnostic_grid.json.

## A.32 Below-threshold exponents

Claim suppression. Source: Ch5 AppendixE/Fig6; PDF114, L8196–8197@P113; also L5997–5998@P94. Published: Suppression beta 1.91(.03) at N0 and 1.15(.03) at N8.

Ours: 0 of 2 suppression fits completed.. Status: not established. 0 of 2 suppression fits completed.

Required scope: Source suppression fits beta at N=0(code capacity) and N=8(single-shot cycles); uncertainties retained separately. No comparable production fits completed.

Evidence: sources/browser-extraction.txt, availability.md, results/original_pilots.json.

## A.33 Random-product single-shot performance

Claim nontopological. Source: Ch5 TableIII/Figure4; PDF95–96, L6015@P94 and L6060–6080@P95. Published: Three random product codes [[1336,4,6]], [[3100,5,8]], [[5964,6,10]]; N16 crossing about 2.7%.

Ours: 0 of 3 performance analyses; original seeds not recovered.. Status: blocked. 0 of 3 performance analyses; original seeds not recovered.

Required scope: Source three random3Dproduct codes [[1336,4,6]],[[3100,5,8]],[[5964,6,10]], with reported approximate crossing near2.7% after16cycles; exact seed inputs unrecovered.

Evidence: sources/browser-extraction.txt, availability.md, results/original_pilots.json.

## A.34 Appendix E thresholds and suppression fits

Claim fit-methods. Source: Dissertation: Appendix E thresholds and suppression fits. Published: Appendix E thresholds and suppression fits.

Ours: Documentary inventory only; no independent proof, resource-count measurement or fit completed for this item. Status: not established. Documentary inventory only; no independent proof, resource-count measurement or fit completed for this item.

Required scope: Implement cited rescaling and extrapolation methods only on adequate comparable data; synthetic recovery tests do not replicate reported fitted values.

Evidence: sources/browser-extraction.txt, availability.md.

## A.35 Limits of a complete HGP computing architecture

Claim limitations. Source: Ch6 PDF119, L8443–8459@P118; Ch5 PDF86 L5051–5056@P85. Published: Chapter6 explicitly leaves a comprehensive framework for encoded operations open. Chapter5 conjectures a stochastic threshold from general

good confinement, while proving the linear-confinement conditional theorem..

Ours: No architecture or conjecture demonstration attempted.. Status: not established. No architecture or conjecture demonstration attempted.

Required scope: Chapter6 open comprehensive encoded-operation framework and Chapter5 conjectural general-confinement stochastic threshold; distinguish established conditional linear-confinement theorem from broader conjecture.

Evidence: sources/browser-extraction.txt, availability.md.

## A.36 Chapter 4 Appendix B Table 2: all nine symmetric codes

Claim symmetric-table. Source: Ch4 Appendix B Table 2. Published: Nine symmetric HGP parameter rows.

Ours: 9 of 9 n,k,seed distance,maxweight rows agree; quantum distance inherited, not exhaustively established. Status: partly replicated. Finite or documentary coverage only; universal and full-design conclusions not independently established. See named artifacts and limitations.

Required scope: Parse all nine printed seed matrices; independently enumerate each classical kernel, construct H transpose H and verify n, k, rate, seed distance and maximum stabilizer weight. Distinguish inherited distance theorem from exhaustive quantum distance search.

Evidence: sources/browser-extraction.txt, availability.md, results/exact_checks.json.

## A.37 Pieceable parallelism and overhead

Claim parallel-overhead. Source: Dissertation: Pieceable parallelism and overhead. Published: Pieceable parallelism and overhead.

Ours: 361 support-size pairs1..19 pass complete pair coverage and collision-free scheduling. Ideal logical action checked for256 Hamming98 left/right pairs. Syndrome-extraction overhead and noisy intermediate correction unmeasured.. Status: partly replicated. Finite result stated explicitly; source-wide generalization or visual wiring not established.

Required scope: Account for Section 4.3 and Figure 7: verify disjoint support criterion and schedule counts; preserve unmeasured syndrome extraction cost and fault-tolerance limits.

Evidence: sources/browser-extraction.txt, availability.md.

## A.38 Random-product code1

Claim random-code1. Source: Ch5 Table III, Figure 4. Published: [[1336,4,6]].

Ours: Required full-design inputs or production estimates not obtained; pilots and inventory are not fitted results.. Status: blocked. Required full-design inputs or production estimates not obtained; pilots and inventory are not fitted results.

Required scope: Locate all three Table III random seed codes and exact Figure 4 grid; validate all parameter entries and run BP+OSD twice for N=16 when inputs permit.

Evidence: sources/browser-extraction.txt, availability.md, results/original_pilots.json.

## A.39 Random-product code2

Claim random-code2. Source: Ch5 Table III, Figure 4. Published: [[3100,5,8]].

Ours: Required full-design inputs or production estimates not obtained; pilots and inventory are not fitted results.. Status: blocked. Required full-design inputs or production estimates not obtained; pilots and inventory are not fitted results.

Required scope: Locate all three Table III random seed codes and exact Figure 4 grid; validate all parameter entries and run BP+OSD twice for N=16 when inputs permit.

Evidence: sources/browser-extraction.txt, availability.md, results/original_pilots.json.

## A.40 Random-product code3

Claim random-code3. Source: Ch5 Table III, Figure 4. Published: [[5964,6,10]].

Ours: Required full-design inputs or production estimates not obtained; pilots and inventory are not fitted results.. Status: blocked. Required full-design inputs or production estimates not obtained; pilots and inventory are not fitted results.

Required scope: Locate all three Table III random seed codes and exact Figure 4 grid; validate all parameter entries and run BP+OSD twice for N=16 when inputs permit.

Evidence: sources/browser-extraction.txt, availability.md, results/original_pilots.json.

### A.41 Random-product crossing

Claim random-crossing. Source: Ch5 Table III, Figure 4. Published: 2.7.

Ours: Required full-design inputs or production estimates not obtained; pilots and inventory are not fitted results.. Status: blocked. Required full-design inputs or production estimates not obtained; pilots and inventory are not fitted results.

Required scope: Locate all three Table III random seed codes and exact Figure 4 grid; validate all parameter entries and run BP+OSD twice for N=16 when inputs permit.

Evidence: sources/browser-extraction.txt, availability.md, results/original_pilots.json.

## B Artifacts and regeneration recipes

The recipes separate rebuilding this report from rerunning scientific calculations and from acquiring source data. Scientific code and inputs were present at repository revision `a18f642710aa192bdaa9f0dc4c68cb32bb08895c`; this is historical run provenance. The wrapper fresh-output fix is revision `005d176ca47b45a901cd8202a2c02071f703e026`. Neither identifier asserts that this manuscript repair was published. Exact dependency pins are in requirements.txt; the complete script recipe is in README.md, subsection Reproduce.

### B.1 Software and engine

The calculation environment used Python 3.12.14, NumPy 2.3.5 for binary arrays, SciPy 1.17.0 for normal quantiles, Matplotlib 3.10.8 for plots, PyYAML 6.0.3 for validation and PyMuPDF 1.26.6 for PDF inspection; CMake 4.4.4 and GNU C/C++ 14.2.0 built the original decoder (Software projects, versioned tools; Single-shot software, archived revision). The report is built with pdfTeX 1.40.26 (TeX Live 2025/dev/Debian) (Software projects, versioned tools). provenance/environment.json and provenance/engine-preflight.json record the environment and accepted engine fallback: exact binary algebra and the original C++ decoder are appropriate to this model; the missing R SEM packages cannot express it. No formal proof assistant certification is claimed.

### B.2 Run calculations

From `formalization/quintavalle-hgp/`, create an isolated Python environment and install `requirements.txt`. Set `OPENBLAS_NUM_THREADS=1` and `MPLCONFIGDIR=/tmp/quintavalle-matplotlib`. Run these exact commands after preserving saved outputs; reruns overwrite result files.

```
python -m venv .venv
. .venv/bin/activate
pip install -r requirements.txt
export OPENBLAS_NUM_THREADS=1
export MPLCONFIGDIR=/tmp/quintavalle-matplotlib
python analysis/verify.py
python analysis/independent_checks.py vendor/single_shot_3D_HGP
python analysis/toric_noise.py
cmake -S vendor/single_shot_3D_HGP \
 -B vendor/single_shot_3D_HGP/build \
 -DCMAKE_POLICY_VERSION_MINIMUM=3.5
cmake --build vendor/single_shot_3D_HGP/build -j2
python analysis/upstream_runs.py vendor/single_shot_3D_HGP
python analysis/upstream_runs.py vendor/single_shot_3D_HGP --suite
```

The toric seed is 20261103; diagnostic seeds are $20261103 + 100i +$ repair for syndrome-noise index $i$, and pilot seeds are $20261103 + i$. Exact enumeration has no seed. Pilot timing and timeout counts may vary. Saved JSON summaries and raw per-run logs provide checkpoints for inspection; the scripts are not represented as resumable production fitting tools. Missing source inputs remain missing after rerunning these commands.

| Claim family | Evidence artifact | Regeneration script |
|---|---|---|
| ReShape, bases, square gates, nine appendix rows, finite confinement | results/exact_checks.json | analysis/verify.py |
| 19 matrices, phases, partitions, round-robin and ideal injection | results/independent_checks.json | analysis/independent_checks.py |
| 56-setting reconstructed toric experiment | results/toric_noise.json | analysis/toric_noise.py |
| Eight original-software diagnostic settings | results/original_diagnostic_grid.json | analysis/upstream_runs.py; use –suite |
| Four feasibility pilots; uncompleted production claims | results/original_pilots.json | analysis/upstream_runs.py; omit –suite |
| All 41 source comparisons and 14 headline bindings | evidence-packet.json; sources/headline-finding-audit.json | Documentary source audit; requires original source for new extraction |
| Source attribution and acquisition record | sources/browser-extraction.txt; availability.md; provenance/source-acquisition.json | Public PDF read; headline pages inspected; file not stored here |
| Binary-algebra dependency | analysis/gf2.py | Imported by scripts above; no standalone target |

### B.3 Rebuild the report from saved outputs

The editable sources are paper/manuscript.tex and paper/references.tex. From `formalization/quintavalle-hgp/paper/`, use the commands below; they do not rerun simulations. Build into a temporary directory and replace the descriptive PDF only after both passes succeed.

```
mkdir -p /tmp/quintavalle-paper-build
pdflatex -interaction=nonstopmode -halt-on-error \
 -output-directory=/tmp/quintavalle-paper-build manuscript.tex
pdflatex -interaction=nonstopmode -halt-on-error \
 -output-directory=/tmp/quintavalle-paper-build manuscript.tex
cp /tmp/quintavalle-paper-build/manuscript.pdf \
 hypergraph_product_codes__finite_support_and_unresolved_production_results.pdf
```

The bibliography uses editable `bibitem` entries in `references.tex`; no BibTeX fetch is needed. Original data acquisition is a separate task from this rebuild.

## C Compliance ledger

The ledger distinguishes fulfilled reporting requirements from intentional model-specific or layout departures.

| Requirement | Status | Evidence and justification |
|---|---|---|
| Verdict and full scope | followed | Abstract, introduction, all 41 inventory items, 14 headline findings and six limitations. |
| Findings separate from effects | followed | Table 1 retains source and finite findings; capacity and suppression effects have separate notes. |

| Requirement | Status | Evidence and justification |
|---|---|---|
| Evidence types | followed | Sections 3–4; no universal proof or noisy architecture inferred. |
| Engine preflight | departed | Accepted model-specific Python/C++ fallback in engine-preflight.json; SEM software cannot represent binary-code algebra. |
| Approved body layout | followed | The manuscript inputs the shared approved preamble: 11-point article, 0.82-inch margins, centered title, and the standard abstract. The 14-row headline table uses a smaller font so it stays on that page. |
| Source page inspection | limited | The thesis PDF was read and the rendered headline pages were inspected; phase conventions and missing grids remain provisional. |
| Citation coverage | followed | Dissertation cited at first attribution; locators throughout inventory. Wilson metadata verified, full scan not text-readable; NIST supplies Bonferroni method. |
| Stochastic uncertainty | followed | Wilson and 56-cell adjusted intervals distinguished from rounding and reconstruction uncertainty. |
| SEM comparability fields | not applicable | No SEM indices, likelihood fits or baseline models are estimated. |
| Reproduction and availability | followed | Appendix B and README list commands, versions, seeds, outputs, source gaps and historical run revisions. |

# References


Quintavalle, Armanda Ottaviano. (2022). *Hypergraph product codes: a bridge to scalable quantum computers*. PhD dissertation, University of Sheffield. https://etheses.whiterose.ac.uk/id/eprint/32300/. Source used: page-marked text extraction; PDF ink unavailable.

Wilson, E. B. (1927). Probable Inference, the Law of Succession, and Statistical Inference. *Journal of the American Statistical Association*, 22(158), 209–212. https://doi.org/10.1080/01621459.1927.10502953. Publisher metadata verified; full scan not independently text-readable in this audit.

NIST/SEMATECH. *e-Handbook of Statistical Methods*, Section 7.4.7.3, Bonferroni's method. https://itl.nist.gov/div898/handbook/prc/section4/prc473.htm. Accessed 3 November 2026.

*single_shot_3D_HGP*. MIT-licensed original software. Revision `bdfb437b2abcfa514997f26be97a711b878448cb`; BP+OSD submodule `74f86d3ef00f04bbb90a043dfef52e92a091f4d3`. https://github.com/MikeVasmer/single_shot_3D_HGP.

Python 3.12.14 (https://github.com/python/cpython); NumPy 2.3.5 (https://github.com/numpy/numpy); SciPy 1.17.0 (https://github.com/scipy/scipy); Matplotlib 3.10.8 (https://github.com/matplotlib/matplotlib); PyYAML 6.0.3 (https://github.com/yaml/pyyaml); PyMuPDF 1.26.6 (https://github.com/pymupdf/PyMuPDF); CMake 4.4.4 (https://github.com/Kitware/CMake); GCC 14.2.0 (https://gcc.gnu.org/); pdfTeX 1.40.26, TeX Live 2025/dev/Debian (https://tug.org/texlive/). Versions recorded in the environment or local build output; links identify software projects.